Vikram Kumar Anand

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EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2027	8.45
SAINIK SCHOOL PURULIA, WEST BENGAL	Class XII (CBSE)	2019	63.20%
SAINIK SCHOOL PURULIA, WEST BENGAL	Class X (CBSE)	2017	9.00%

SCHOLASTIC ACHIEVEMENTS

- **All India Rank 39** in **SSC (Tech)-67** course for entry into the Indian Army; successfully cleared the **SSB Interview at 14 SSB, Prayagraj**.
- **Qualified UPSC Combined Defence Services (CDS) Examination**, demonstrating strong aptitude, analytical ability, and leadership potential.
- **Qualified AFCAT (Air Force Common Admission Test)** conducted for selection into the Indian Air Force.

POSITIONS OF RESPONSIBILITY

Class Representative (CR), MAE Department, IIT Hyderabad

- Represent the student community of the **Mechanical & Aerospace Engineering (MAE) department**, acting as a communication bridge between students, faculty, and departmental administration .Coordinate academic and departmental communication, collect student feedback, and communicate student concerns and requirements to the concerned authorities.

SKILLS

- **Programming:** Python, C++
- **Robotics & Autonomous Systems:**ROS 2, SLAM, Autonomous Navigation, Robotics, Sensor Fusion
- **Computer Vision & AI:**Computer Vision, Machine Learning, Image Processing
- **Simulation & Engineering Tools:**MATLAB, Fusion 360
- **Embedded Systems & Hardware:**Arduino, Sensors & Actuators
- **Control & Automation:**Control Systems, Robotics & Automation

RELEVANT COURSES

- Machine Learning
- Fundamentals of Robotics
- Marine Robotics
- Computer Vision
- Autonomous Robotic Systems
- Soft Robotics
- Probability & Optimization
- Control System Design
- Sensor & Actuator Laboratory
- Robotics & Automation Laboratory

PROJECTS

1. Design and Development of an Energy-Aware Collaborative Multi-USV System with Autonomous Docking and Wireless Recharging

M.Tech Thesis | Ongoing

- Developing a **real-world collaborative multi-USV system** integrating autonomous navigation, VSLAM-based localization, inter-USV communication, and energy-aware mission coordination.
- Integrating **Side-Scan Sonar** for detection and localization of underwater sludge, man-made waste, and seabed anomalies.
- Developing and validating **autonomous docking and wireless recharging** with a mothership, progressing from simulation to **working real-world deployment**.

2. Autonomous Navigation of a Mobile Robot

- Developed a **ROS 2-based autonomous navigation system** using LiDAR, SLAM, localization, waypoint navigation, and path planning.
- Implemented **real-time obstacle avoidance** and autonomous navigation in unknown environments.
- Validated the complete navigation system in **both simulation and on a physical mobile robot**.

3. Design-Oriented Workspace Analysis and Pick-and-Place Simulation

- Designed and modeled a robotic manipulator using **MATLAB Robotics System Toolbox and URDF**, with workspace and kinematic analysis.
- Evaluated **reachable workspace and design modifications**, including link/joint parameter effects.
- Implemented **obstacle-aware path and trajectory planning** for autonomous pick-and-place simulation.

4. GPS-Based Waypoint Navigation of a Quadcopter

- Developed a **GPS-based autonomous waypoint navigation system** for a quadcopter.
- Implemented waypoint generation, sequencing, GPS-based localization, and autonomous position navigation.
- Integrated high-level navigation commands with the **flight-control system** for autonomous mission execution.

6. Buoyancy-Driven Underwater Glider

- Designed and developed a **buoyancy-driven underwater vehicle** using a syringe-based variable-buoyancy mechanism, lead screw, stepper motor, and Arduino.
- Implemented automated **water intake/expulsion and buoyancy control** using motor-driven actuation and limit switches.
- Built and experimentally validated the prototype for **controlled underwater motion**.

Computer Vision-Based Image Classification Using Deep Learning

Python, PyTorch, Torchvision

Built and trained three neural network models, including a CNN, to classify clothing images from the Fashion-MNIST dataset. Compared their accuracy, loss, and training time, with the CNN achieving **88.52% test accuracy**. Also implemented prediction visualisation, confusion matrix analysis, and model saving and reloading to evaluate the trained model.

GPT-Based Large Language Model (LLM) from Scratch

Python, PyTorch, tiktoken

Implemented a GPT-style language model from scratch using PyTorch, covering Transformer architecture, self-attention, and next-token prediction. Trained the model on text data and experimented with text generation. Also loaded pretrained GPT-2 weights into the custom architecture and explored text generation using temperature scaling and top-k sampling.

AI-Powered Indian Banking Research Assistant Using LLMs, RAG, and Financial Analytics

- Developed a financial research assistant using Qwen2.5-0.5B-Instruct to answer questions from RBI, HDFC Bank, and ICICI Bank reports.
- Implemented hybrid document retrieval using Sentence Transformers, FAISS, and BM25 to improve evidence retrieval and source-based question answering.
- Integrated Python-based financial analytics using yfinance, Pandas, and NumPy to calculate historical stock returns and volatility, with a Gradio interface for interactive financial research.

EXTRACURRICULAR

1. **Hockey -- Zonal Player**, representing/competing at the zonal level and demonstrating teamwork, discipline, and competitive spirit.
2. **NCC -- B Certificate Holder**, with training and experience in discipline, leadership, teamwork, and organizational activities.
3. **Basketball**, actively participated in the sport, developing teamwork and coordination.
4. **Badminton**, actively participated in recreational/competitive play.

5. **Trekking**, pursuing outdoor activities and endurance-based challenges.
6. **Cycling**, regular recreational and fitness activity.
7. **Cooking**, personal interest and hobby.
8. **Anchoring**

EXPERIENCE

Robotics Engineering Intern | Jingu's Labs Private Limited, Hyderabad

Nov 2025 - Dec 2025

- Designed the **3D mechanical model of a gantry-based Smart Kitchen system** using **Fusion 360**, focusing on the overall robotic architecture and component integration.
- Modeled and integrated **linear actuators, NEMA 16 stepper motors, sensors, camera, and utensil-manipulation end-effector** within the gantry system.
- Contributed to the mechanical design of an **automated cooking platform**, translating the functional requirements into a practical 3D robotic system.

PUBLICATIONS

Co-authored and published a conference paper at **IMECE (International Mechanical Engineering Congress and Exposition)** titled *"Hierarchical Control of Cooperative Quadrotor Payload Transport."* The work investigates **hierarchical control and cooperative autonomous aerial robotics for coordinated payload transportation using multiple quadrotors.**